Project Report

Title: Cloud & DevOps Engineering Projects Report
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Course: Cloud Computing & DevOps
Institute: Besant Technologies, Bengaluru
Duration: Jun 2025 – Present
Role: Cloud & DevOps Trainee

Declaration
I hereby declare that this project report titled **"Cloud & DevOps Engineering Projects"** is my original work carried out during my training period at Besant Technologies. This work has not been submitted previously for any academic qualification. All references and resources used are duly acknowledged.

Acknowledgement
I would like to express my sincere gratitude to Besant Technologies and my trainers for their continuous guidance and support throughout my Cloud & DevOps training. I also thank my peers for their collaboration and encouragement, which helped me successfully complete these projects.

Abstract
This report presents a comprehensive overview of multiple Cloud and DevOps projects implemented using AWS, Azure, Linux, networking concepts, and modern DevOps tools. The projects focus on automating infrastructure provisioning, implementing CI/CD pipelines, deploying scalable cloud-native applications, and improving system reliability and availability. These implementations demonstrate realworld DevOps practices such as Infrastructure as Code (IaC), containerization, orchestration and automation.

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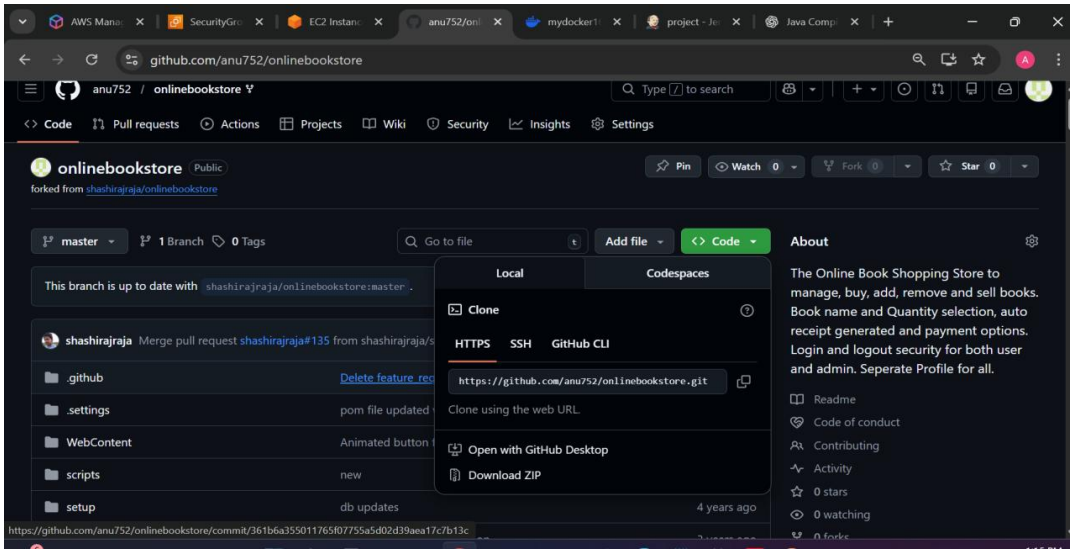
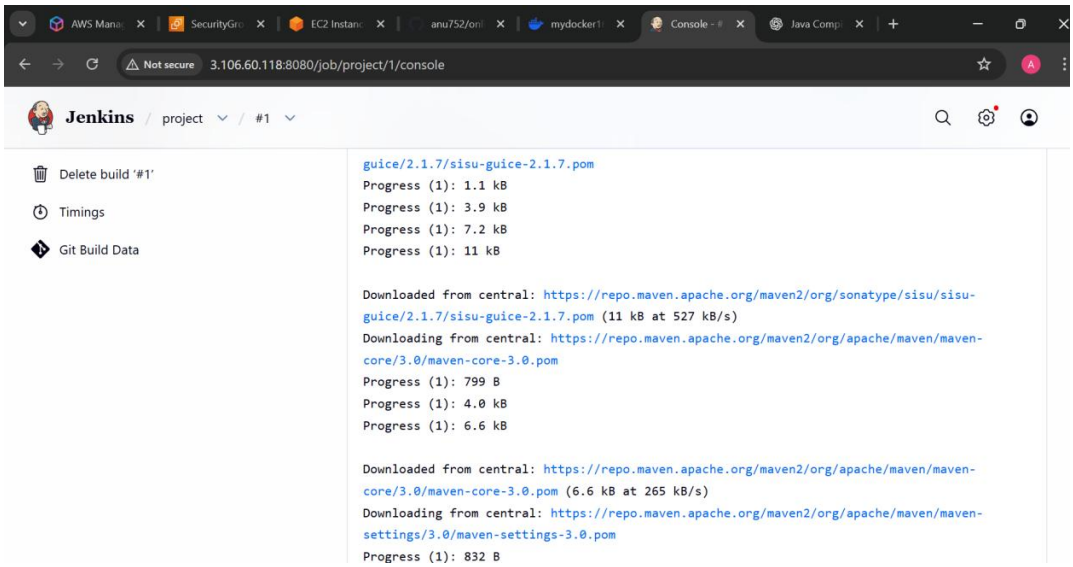
Introduction
Cloud computing and DevOps have become essential for delivering scalable, reliable, and fast software solutions. This project report showcases hands-on implementations that combine cloud infrastructure, automation, and CI/CD practices. The objective is to reduce manual effort, improve deployment speed, and ensure high availability and security of applications.

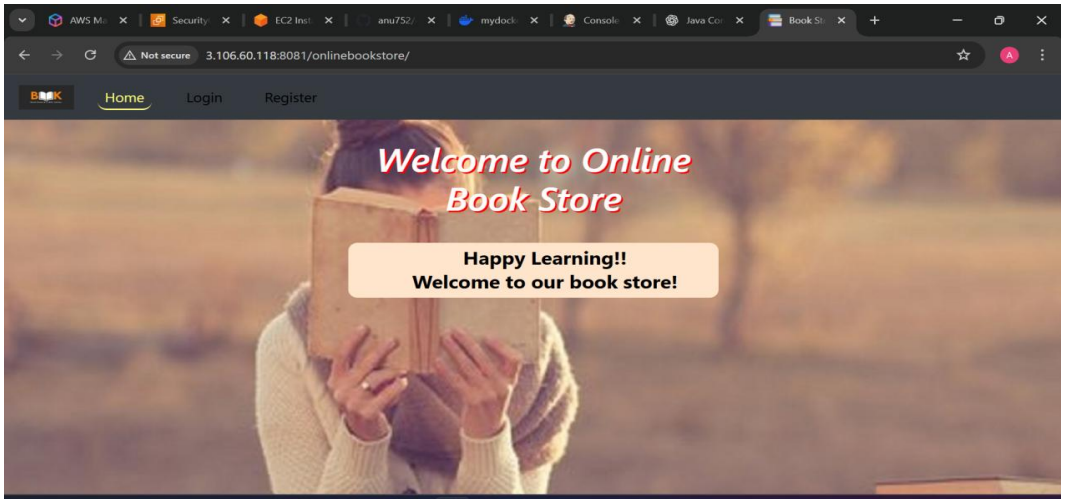
Tools & Technologies Used
• **Cloud Platforms:** AWS
• **Operating Systems:** Linux (RedHat, Ubuntu)
• **DevOps Tools:** Git, GitHub, Jenkins, Docker, Kubernetes
• **Programming & Scripting:** Python, Shell Scripting
• **Networking:** VPC, Subnets, Routing, Load Balancers, Security Groups

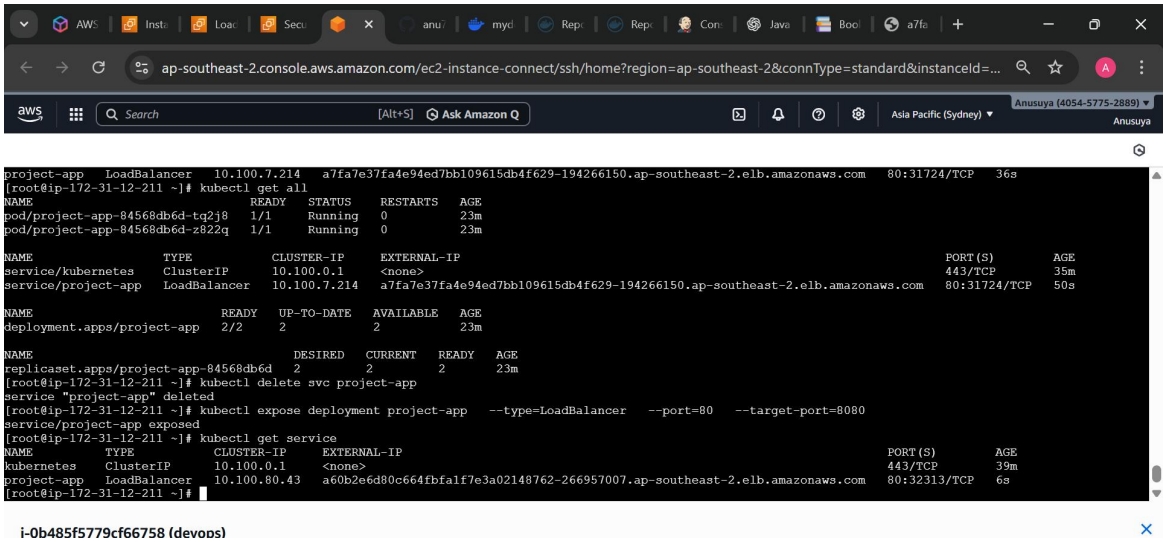
Projects
Project – 1 : Automated Cloud-Native CI/CD Pipeline

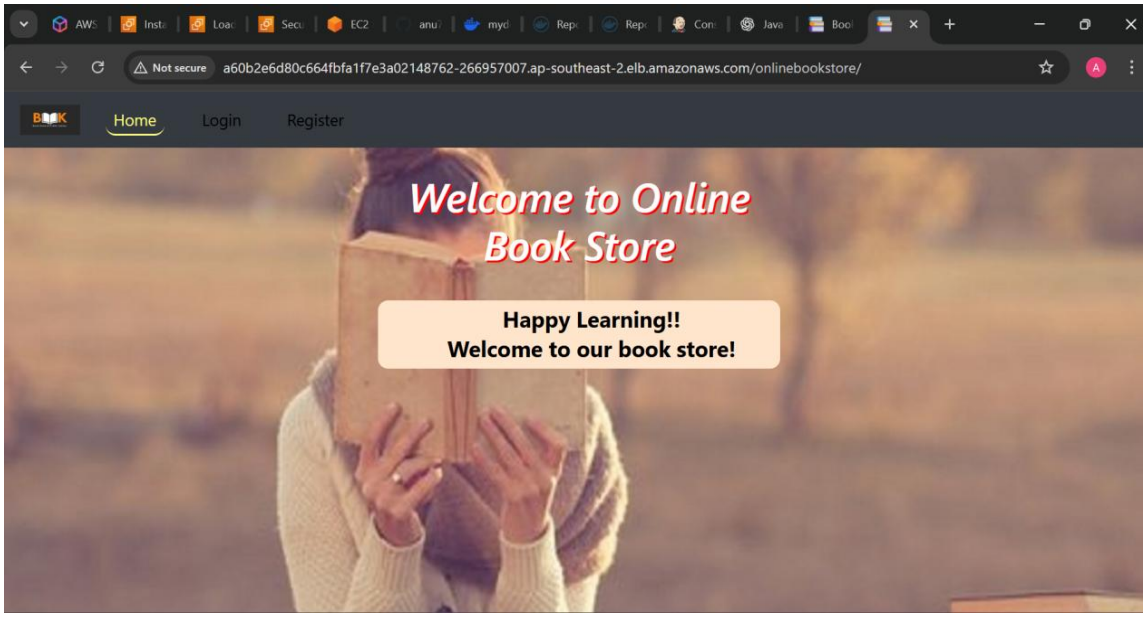
This project focuses on building an automated CI/CD pipeline to deploy cloud-native applications in a consistent and reliable manner. Source code was managed using GitHub, and Jenkins was used to automate build and deployment processes. Docker was used to containerize the application, and Kubernetes handled deployment and scalability on AWS infrastructure.
Tools Used: GitHub, Jenkins, Docker, Kubernetes, AWS









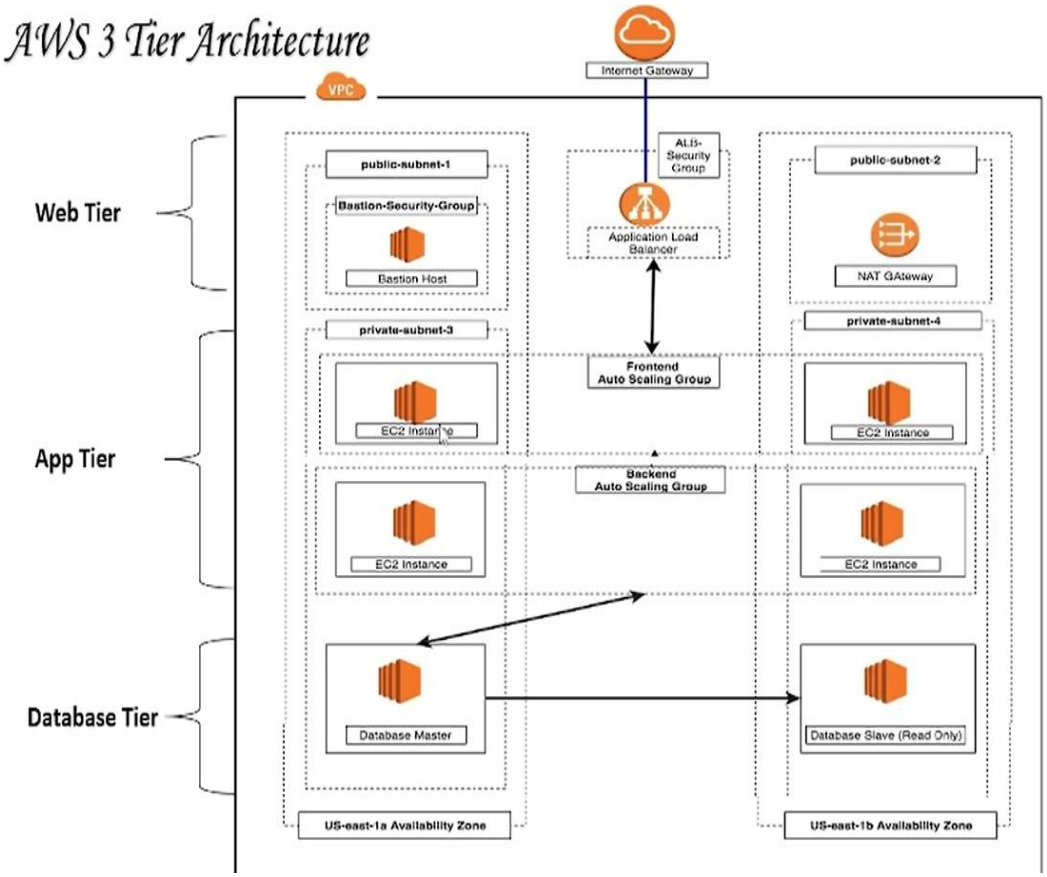


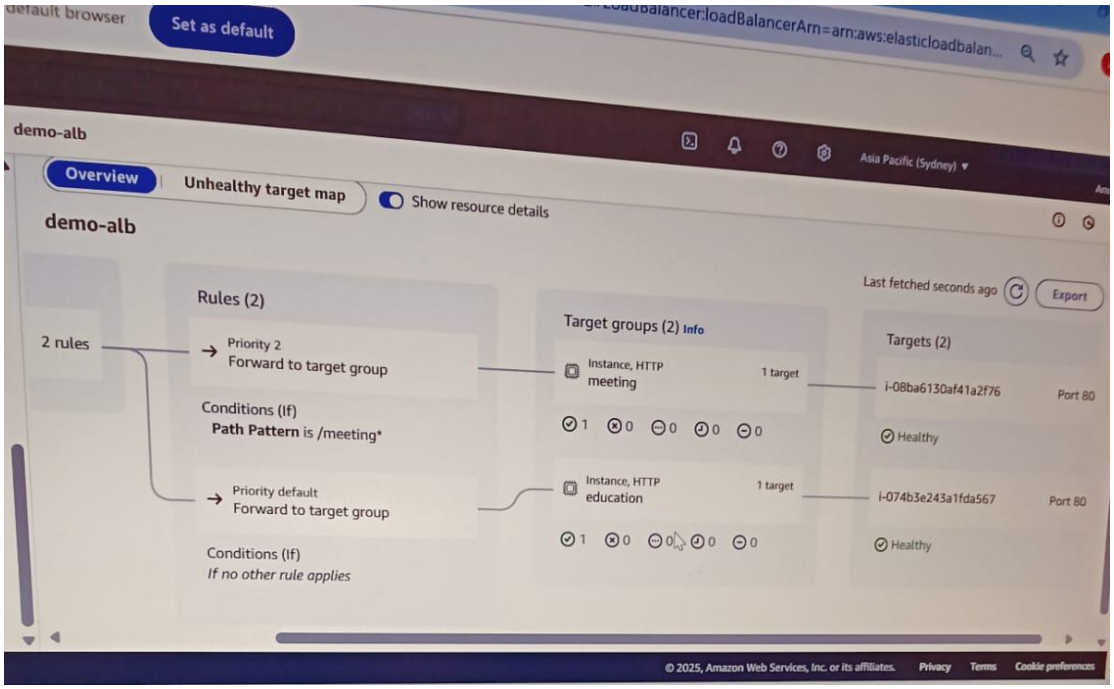
Outcome:
Improved deployment speed, reduced manual effort, and achieved reliable and scalable application delivery.

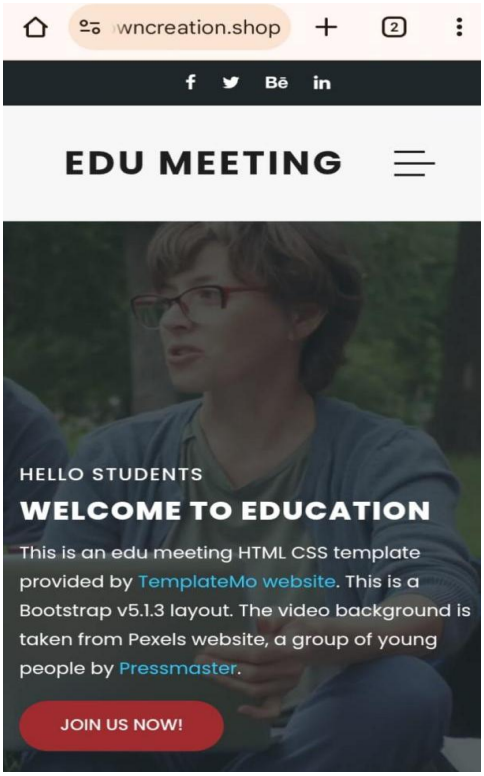
Project – 2: AWS 3-Tier Architecture Implementation

In this project, a secure and scalable 3-tier architecture was designed and implemented on AWS. The architecture separates web, application, and database layers using public and private subnets. Load Balancers and Auto Scaling ensured high availability, while security groups and IAM improved overall security.

Tools Used: AWS EC2, VPC, ALB, Auto Scaling, RDS, IAM, Route53


AWS 3 Tier Architecture
Internet Gateway
VPC
Web Tier
App Tier
Database Tier
public-subnet-1
Bastion-Security-Group
Bastion Host
ALB-Security Group
Application Load Balancer
public-subnet-2
NAT GAteway
private-subnet-3
private-subnet-4
Frontend Auto Scaling Group
EC2 Instance
Backend Auto Scaling Group
Database Master
Database Slave (Read Only)
US-east-1a Availability Zone
US-east-1b Availability Zone


Set as default
demo-alb
Overview
Unhealthy target map
Show resource details
Asia Pacific (Sydney)
Last fetched seconds ago
Export
2 rules
Rules (2)
Priority 2
Forward to target group
Conditions (If)
Path Pattern is /meeting*
Priority default
Forward to target group
Conditions (If)
If no other rule applies
Target groups (2) Info
Instance, HTTP
meeting
1 target
Instance, HTTP
education
1 target
Targets (2)
i-08ba6130af41a2f76
Port 80
Healthy
i-074b3e243a1fda567
Port 80
Healthy
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



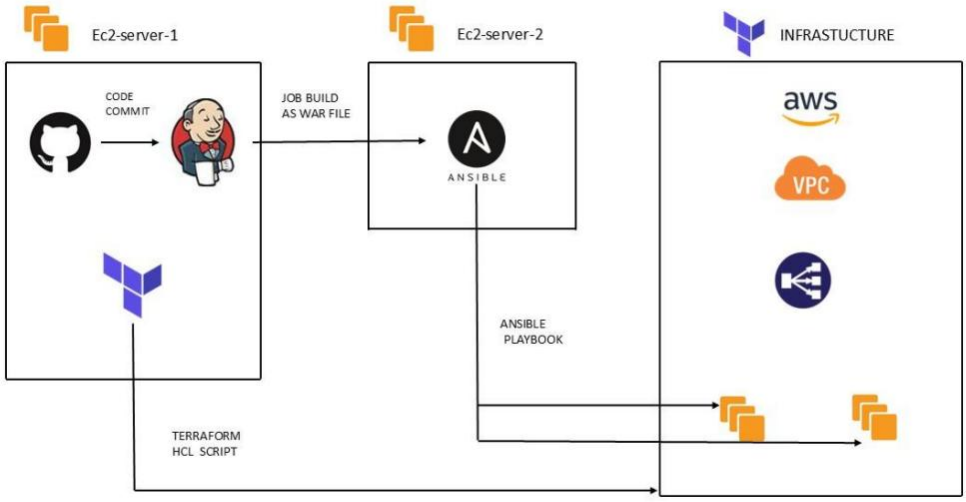
Outcome:

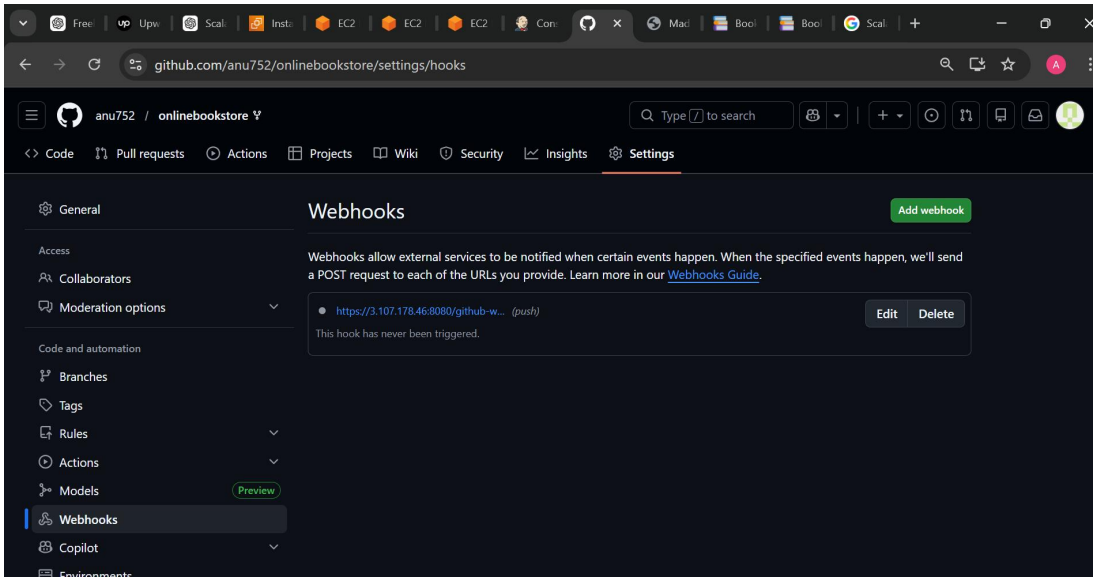
Achieved high availability, improved security, and better performance through layered architecture design.

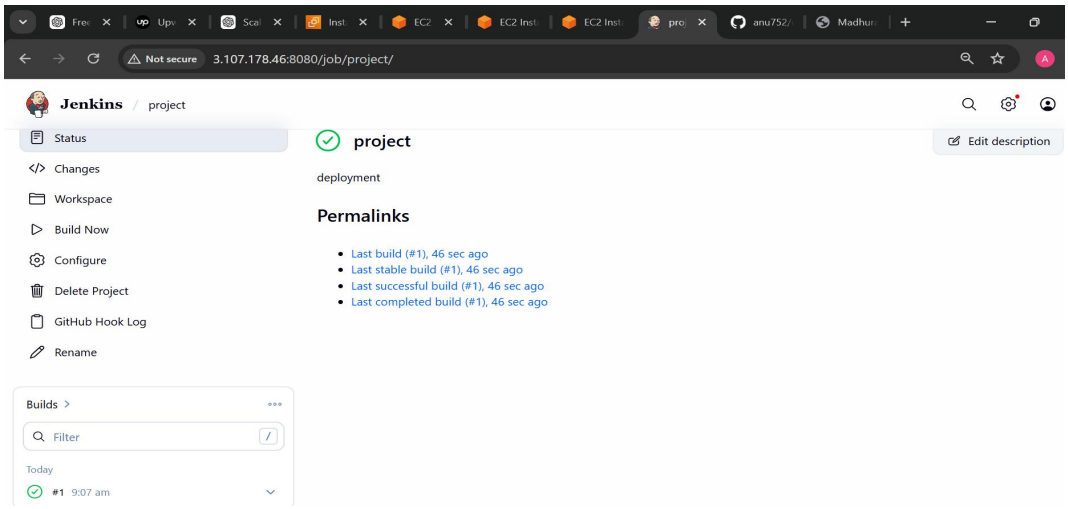
Project – 3 : Scalable Web App Deployment using Terraform & Ansible

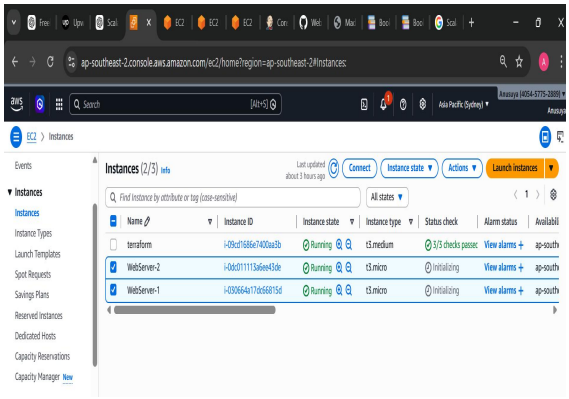
This project demonstrates the use of Infrastructure as Code and configuration management to deploy a scalable web application on AWS. Terraform was used to provision cloud resources such as VPC, EC2 instances, and Load Balancer. Jenkins handled CI/CD automation, while Ansible was used to configure servers and deploy the application automatically.

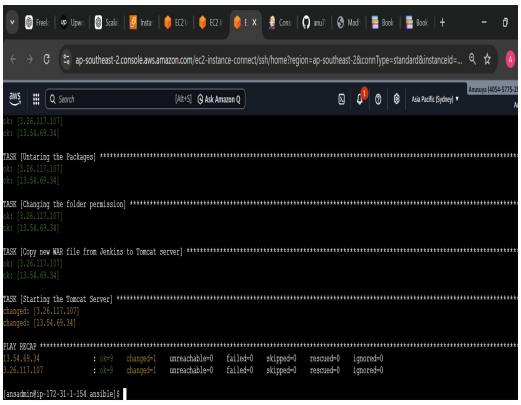
Tools Used: AWS, Terraform, Jenkins, Ansible, GitHub, Linux

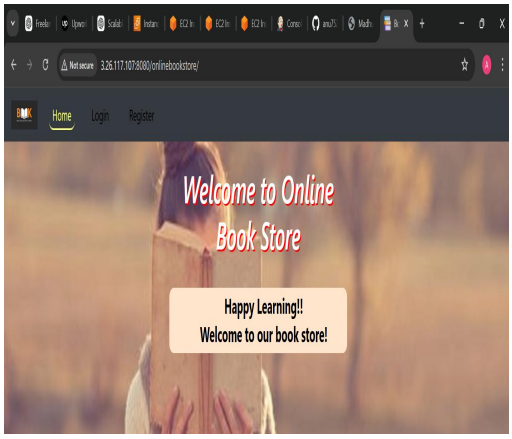


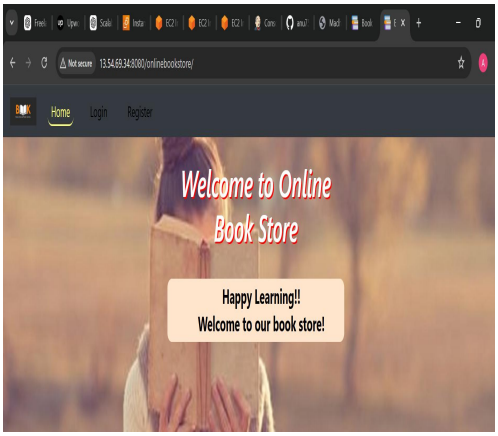

github.com/anu752/onlinebookstore/settings/hooks
anu752 / onlinebookstore
Code
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
Webhooks
Add webhook
Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our Webhooks Guide.
https://3.107.178.46:8080/github-w... (push)
This hook has never been triggered.
Edit
Delete


Not secure
3.107.178.46:8080/job/project/
Jenkins / project
Status
Changes
Workspace
Build Now
Configure
Delete Project
GitHub Hook Log
Rename
Builds
Filter
Today
#1 9:07 am
project
Edit description
deployment
Permalinks
Last build (#1), 46 sec ago
Last stable build (#1), 46 sec ago
Last successful build (#1), 46 sec ago
Last completed build (#1), 46 sec ago


Instances (2/3)
Connect
Instance state
Actions
Launch instances
terraform
WebServer-2
WebServer-1





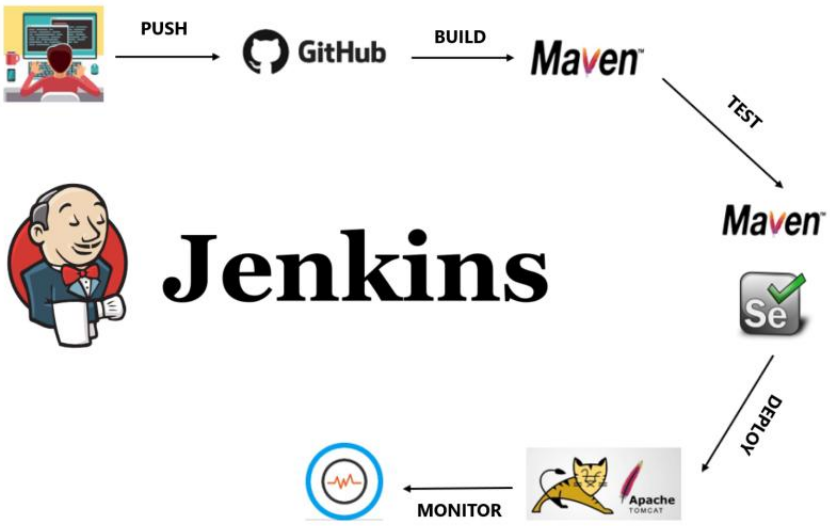


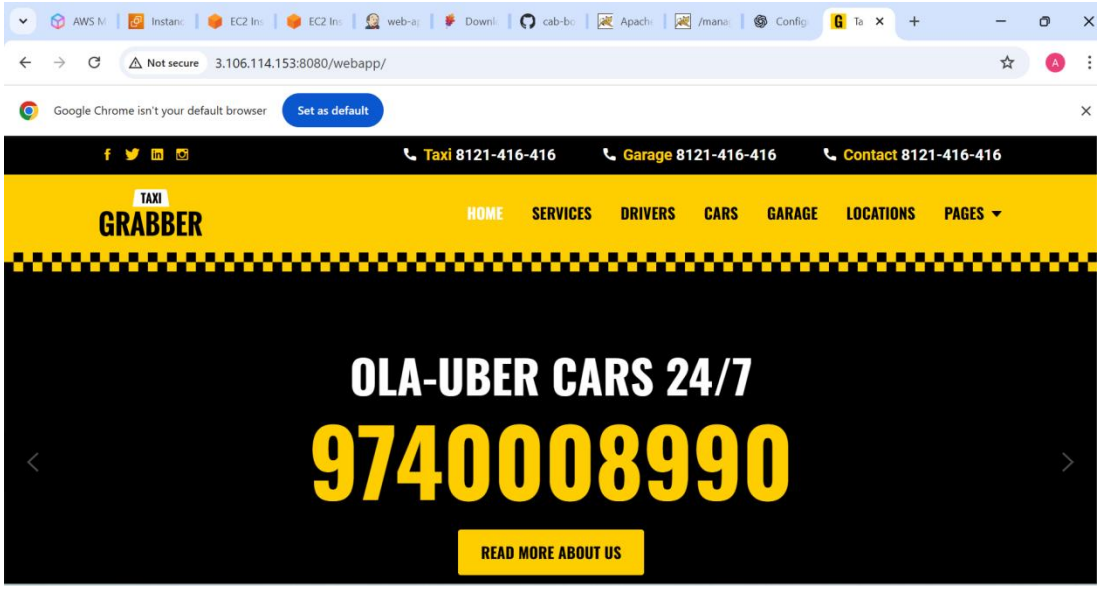
Outcome:
Enabled repeatable infrastructure provisioning and automated application deployment with minimal manual intervention.

Project- 4 : Jenkins-Based CI/CD Pipeline for Java Web Application

This project focuses on implementing an automated CI/CD pipeline for a Java-based web application using Jenkins. The pipeline triggers automatically when code is pushed to GitHub, builds and tests the application using Maven and Selenium, and deploys the application to an Apache Tomcat server.

Tools Used: GitHub, Jenkins, Maven, Selenium,Apache Tomcat, Linux



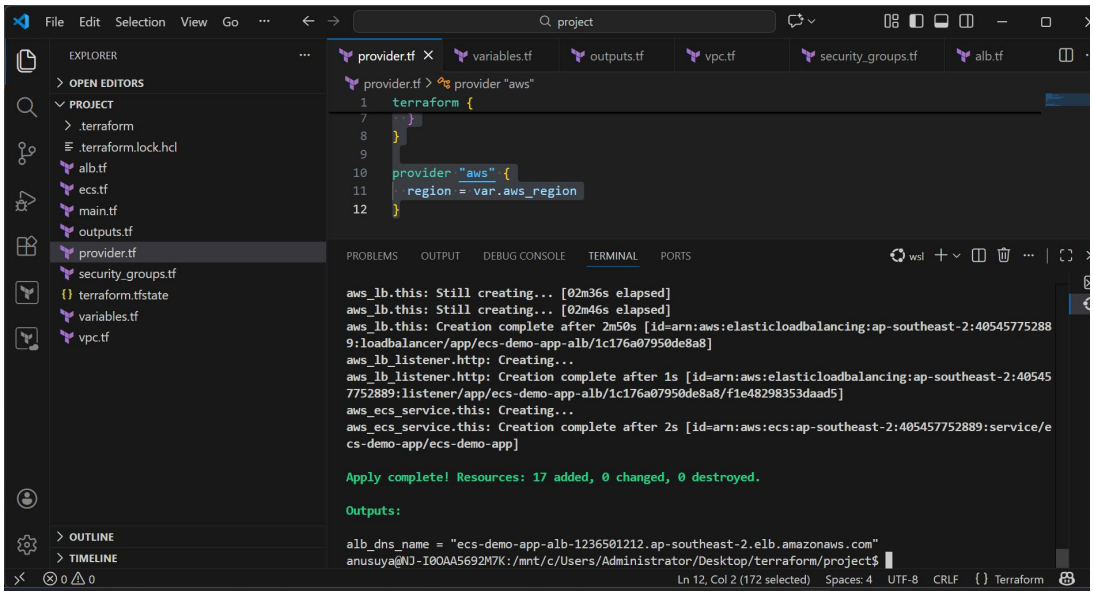


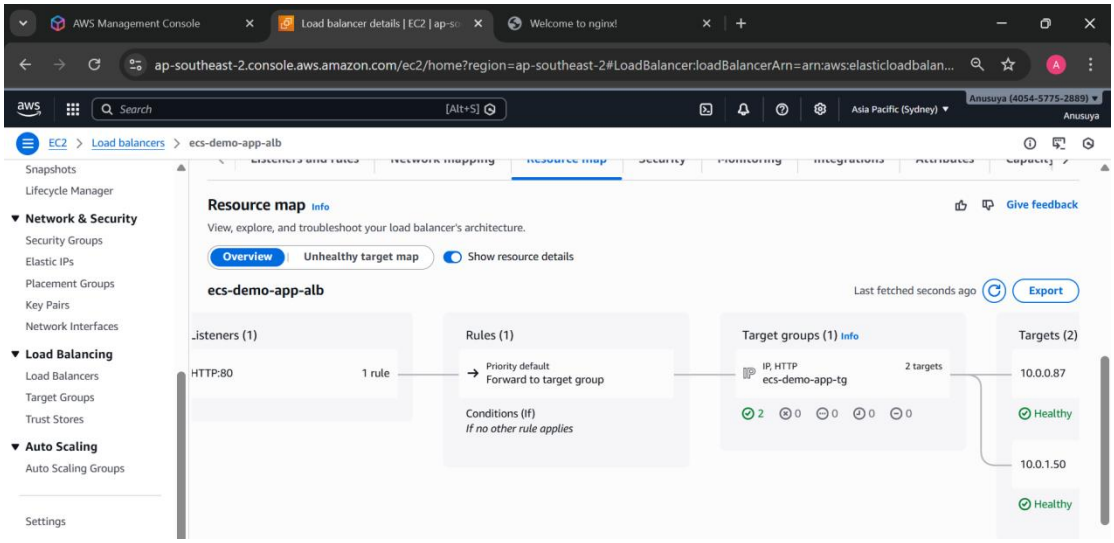
Outcome:
Automated build, test, and deployment process, reduced manual intervention, and improved reliability and delivery speed of the application.

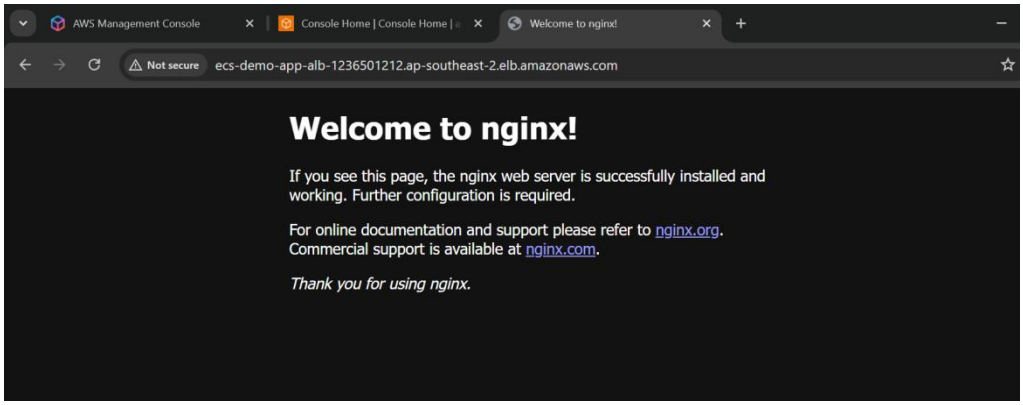
Project 5: Terraform — Deploy Dockerized App on AWS ECS with Load Balancer

This project demonstrates how to deploy a Dockerised app on Amazon ECS. Amazon ECS is the AWS service to run Docker applications on a scalable cluster. It is a container orchestration/ management service similar to Kubernetes

Tools Used: AWS EC2, ELB, IAM, ECS, VPC, Terraform, vs code







Outcome:

This project demonstrated the process of deploying a Dockerized application on Amazon ECS using Terraform. It covered setting up ECS resources, and configuring networking.we can automate the deployment of containerized apps on ECS for scalable and reliable infrastructure on AWS.

Learning Outcomes

- Practical exposure to real-world DevOps workflows
- Strong understanding of cloud infrastructure and automation
- Improved troubleshooting and problem-solving skills
- Hands-on experience with CI/CD, IaC, and monitoring tools

Conclusion

These projects strengthened my understanding of Cloud and DevOps principles and provided hands-on experience in designing, deploying, and automating modern cloud infrastructures. The work aligns with industry standards and prepares me for entry-level Cloud or DevOps roles.